Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits
CSE325	OPERATING SYSTEMS LABORATORY	0	0	2	1
Course Weightage	ATT: 5 CAP: 45 ETP: 50	**Exam Category: X6: Mid Term Exam: Not Applicable – End Term Exam: Practical**			
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP				

Course Outcomes :Through this course students should be able to

CO1 :: execute Linux commands and basic shell programming tasks for script development

CO2 :: identify and describe the role of environment variables, process image replacement in Linux systems

CO3 :: utilize file system-related system calls to manage files efficiently.

CO4 :: demonstrate the creation and management of processes and threads using Linux system calls

CO5 :: illustrate the use of synchronization mechanisms like mutexes and semaphores to prevent race conditions

CO6 :: explain and implement inter-process communication techniques such as pipes, shared memory, and message passing

	TextBooks (T)		
Sr No	**Title**	**Author**	**Publisher Name**
T-1	BEGINING LINUX PROGRAMMING	NEIL MATHEW & RICHARD STONES	WILEY

	Reference Books (R)		
Sr No	**Title**	**Author**	**Publisher Name**
R-1	OPERATING SYSTEM CONCEPTS	ABRAHAM SILBERSCHATZ, GALVIN	WILEY
R-2	UNIX CONCEPTS AND APPLICATIONS	SUMITABHA DAS	Tata McGraw Hill, India

Relevant Websites (RW)		
Sr No	**(Web address) (only if relevant to the course)**	**Salient Features**
RW-1	https://www.elprocus.com/inter-process-communication/	Inter Process Communication
RW-2	http://www.yolinux.com/TUTORIALS/LinuxTutorialPosixThreads.html	Thread Creation
RW-3	https://dextutor.com/fork-system-call/	process creation

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RW-4	https://dextutor.com/what-is-shell-scripting/	Shell programming
RW-5	https://dextutor.com/write-read-system-call/	System calls

Audio Visual Aids (AV)		
Sr No	**(AV aids) (only if relevant to the course)**	**Salient Features**
AV-1	https://youtu.be/rFwdniyo0CM	basic commands to begin Linux
AV-2	https://youtu.be/afmzgInow58	Lab Exp 01: Introduction to Linux Commands || Operating Systems Lab
AV-3	https://youtu.be/8i5B9QZLB_M	Lab Exp 02:: Shell Programming || Session 01: Directory Hierarchy & Types of Commands
AV-4	https://www.youtube.com/watch?v=M-Eo8QkG2gI&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=10	Lab Exp 03:: File Manipulation using System Calls
AV-5	https://www.youtube.com/watch?v=MEqS5KqC_lM&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=15	Lab Exp 04:: Directory Creation using mkdir() System Call
AV-6	https://www.youtube.com/watch?v=_URa9ndc4YE&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=17	Lab Exp 05:: fork system call || getpid and getppid system call || processes creation using fork
AV-7	https://www.youtube.com/watch?v=GP16vv0iEBM&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=21	Lab Exp 06:: Creation of Multithreaded Processes using Pthread Library in C programming
AV-8	https://www.youtube.com/watch?v=86zblXC1tEA&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=24	Lab Exp07:: How to use Mutex to Synchronize Threads in C programming?
AV-9	https://www.youtube.com/watch?v=0UTb6w_pWpo&list=PLWjmN065fOfGdAZrlP6316HVHh8jlvefD&index=28	Lab Exp08:: Interprocess Communication using pipe() || Un-named pipe
AV-10	https://www.youtube.com/watch?v=gr7oaiUsxSU	RPC Vs Simple Procedure Call - Georgia Tech - Advanced Operating Systems

Scheme for CA:

CA Category of this Course Code is:A0202 (2 out of 2)

Component	**Weightage (%)**	**Mapped CO(s)**
Recoup CA		

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Recoup CA	subjective test based on the syllabus	Deactivated 10June 25Additional chance of CA	Individual	Online	30	8 / 10

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Introduction to Linux	Basic Linux Commands: ls, cat, man, cd, touch, cp, mv, rmdir, mkdir, rm, chmod, pwd		Introduction to Linux. Students will be able to understand basic commands of Linux
	Introduction to Linux	System Calls: Read, Write, Open		Introduction to Linux. Students will be able to understand basic commands of Linux
	Introduction to Linux	Lseek		Introduction to Linux. Students will be able to understand basic commands of Linux
Practical 2	shell programming	variables		Students learn to work with variables, i/o redirection and will be able to perform arithmetic operations
	shell programming	standard input/output redirection		Students learn to work with variables, i/o redirection and will be able to perform arithmetic operations
	shell programming	shell arithmetic		Students learn to work with variables, i/o redirection and will be able to perform arithmetic operations. In practical 3, the lab evaluation 1 will be conducted.
Practical 3	shell programming	shell arithmetic		Students learn to work with variables, i/o redirection and will be able to perform arithmetic operations. In practical 3, the lab evaluation 1 will be conducted.
Practical 4	shell programming	flow control and decision making		Students will be able to implement shell programs with flow control and looping statements
Practical 5	File and directory management using system calls	File related system calls (open, read, write, lseek, close)		Students learn to manipulate files using file system calls. Lab Evaluation -2 will be conducted.
Practical 6	File and directory management using system calls	File related system calls (open, read, write, lseek, close)		Students learn to manipulate files using file system calls. Lab Evaluation -2 will be conducted.
Practical 7	File and directory management using system calls	Directory related system calls (opendir, readdir, closedir etc)		Students learn to work with directories using directory system calls
Practical 8	Process creation and threading	Process duplication using fork()		Students will be able to write programs to duplicate processes

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Practical 8	Process creation and threading	Creating processes		Students will be able to write programs to duplicate processes. Lab Evaluation -3 will be conducted.
	Process creation and threading	Replacing process image using execlp		Students will be able to write programs to duplicate processes. Lab Evaluation -3 will be conducted.
Practical 9	Process creation and threading	Replacing process image using execlp		Students will be able to write programs to duplicate processes. Lab Evaluation -3 will be conducted.
	Process creation and threading	Creating processes		Students will be able to write programs to duplicate processes. Lab Evaluation -3 will be conducted.
Practical 10	Process creation and threading	Creating threads using pthread		Students will be able to write programs to create Threads
	Process creation and threading	Creating Threads		Students will be able to write programs to create Threads
	Process creation and threading	Environment variables		Students will be able to write programs to create Threads
Practical 11	Synchronization	Synchronization with Mutexes		Students learn to implement solutions to synchronization problems
	Synchronization	Synchronization with semaphores		Students learn to implement solutions to synchronization problems
	Synchronization	Race Condition		Students learn to implement solutions to synchronization problems. Lab Evaluation -4 will be conducted.
Practical 12	Synchronization	Race Condition		Students learn to implement solutions to synchronization problems. Lab Evaluation -4 will be conducted.
Practical 13	Inter-process communication	Pipes, popen and pclose functions		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Stream pipes, passing file descriptors		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Shared memory		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Message passing		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Remote Procedure calls		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
Practical 14	Inter-process communication	Remote Procedure calls		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Message passing		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications

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Practical 14	Inter-process communication	Shared memory		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Stream pipes, passing file descriptors		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	Inter-process communication	Pipes, popen and pclose functions		Students learn to create processes that can share data with each other and able to implement IPC using RPC between client and server applications
	SPILL OVER			
Practical 15	Spill Over			

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